Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Operationally flexible (80–150 °C) high temperature proton exchange membrane fuel cells (HT-PEMFCs) with no external humidification			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New/renewable energy	Fuel cell – PEMFC	High polymer materials	Material technology for energy and environmental industry purpose
Concept of Proposed Technology	**Concept and Background** Among alternative energy solutions, hydrogen has been recognized as an attractive clean fuel because of its high gravimetric energy density and zero CO_2 emission capability. Among hydrogen fuel cell types, proton exchange membrane (PEM) fuel cell is the most advanced one. In PEM fuel cells (PEMFCs), a PEM is a crucial component, allowing proton transport between two electrodes while preventing diffusion of H_2 and O_2. Currently, commercial PEMFCs use a perfluorosulfonic acid (PFSA) ionomer, such as Nafion, which operates only at high relative humidity (RH) and temperatures below 100 °C (i.e., low-temperature PEMFCs as shown in Figure 1 and Figure 2a). Such low-temperature PEMFCs require humidified hydrogen feed because the sulfonate sidechain of PFSA needs to be solvated by water to provide high proton conductivity. Therefore, the			

development of highly conductive PEM materials that do not need external humidification is highly desirable as they can eliminate a radiator to release heat and significantly simplify the water management (see high-temperature PEMFCs in Figure 1). Furthermore, the capability of high-temperature operation effectively addresses heat dissipation challenges by increasing the temperature gradient, thereby eliminating the need for bulky cooling systems. This advantage allows for the versatile integration of HT-PEMFCs into a wide range of high-power, compact applications where space and weight efficiency are critical. [1-3]

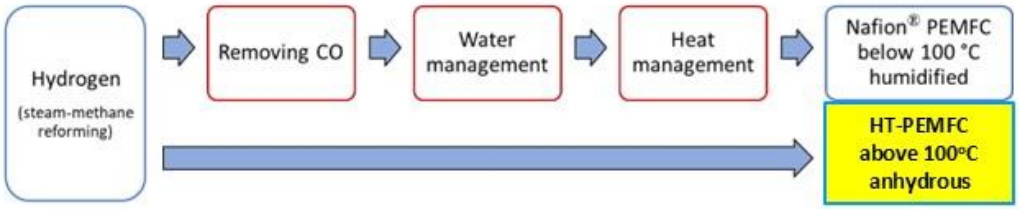

Figure 1. Comparison of low-temperature PEMFC vs. high-temperature PEMFC.

Core technical gap (problem statement)

Phosphoric acid (PA)-doped polybenzimidazole (PA-PBI) membranes have been reported to offer good proton conductivity without external humidification at high temperatures in the range of 140–180 °C. [4] However, PA-PBI maintain good proton conductivity only under anhydrous conditions (i.e., less than 10% RH). If moisture is present to the above that level, severe performance degradation can take place as a consequence of the leaching of the doped phosphoric acid from the PBI membrane. In other words, PA-PBI PEM has very narrow RH tolerance (see Figure 2b). To fill the operation gap between Nafion (low temperature and high RH; below 100 °C and above 80% RH) and PA-PBI (high temperature and anhydrous: at 140–180 °C and less than 10% RH), a PA-doped polyphenylene-type ion-pair coordinated PEM, QAPOH, has been reported to show stable proton conductivity up to 40% RH at 80 °C. [5] The enhanced RH tolerance of QAPOH was attributed to the strong ion-pair interaction between positively charged quaternary ammonium (QA) cations and negatively charged $H_2PO_4^-$ (dihydrogen phosphate) anions. The strong electrostatic force of QA-phosphate ion-pair interaction impeded leaching of doped PA from the membrane at high RH. Despite the promising fuel cell performance report, the commercialization feasibility of ion-pair HT-PEMFCs have been limited due to the complex synthesis and scale-up difficulty of QAPOH.

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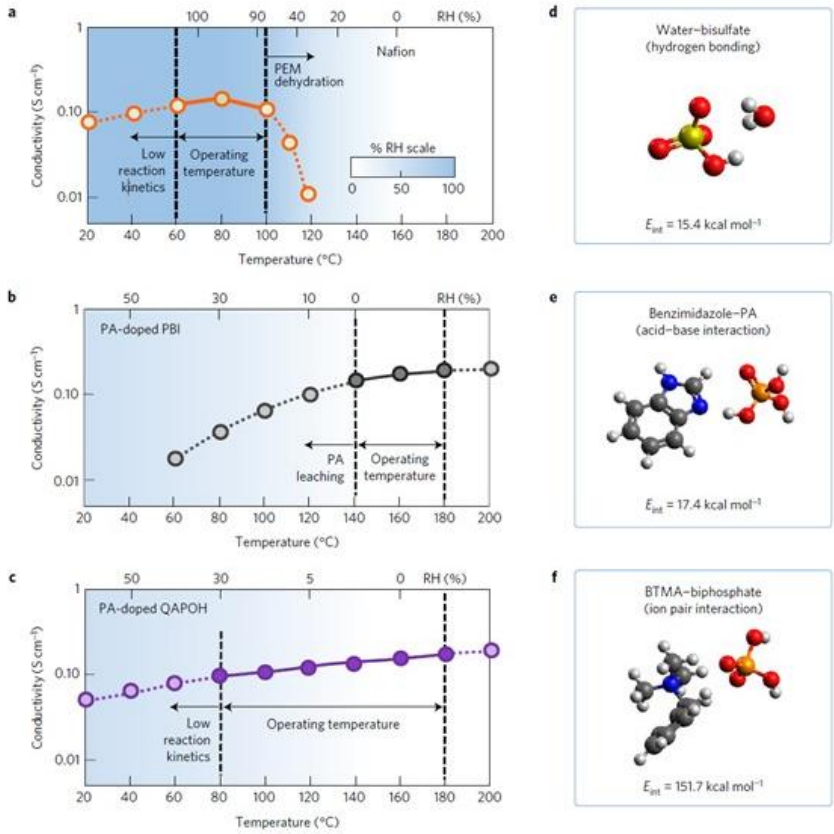

Figure 2. Graphical illustration of proton conductivity pattern as a function of temperature and relative humidity: (a) Nafion, (b) phosphoric acid-doped PBI, (c) phosphoric acid-doped ion-pair HT-PEM (from reference 5).

RPI Innovation

Bae group of RPI has developed a highly ion conductive, chemically and mechanically stable, easily manufacturable ion-pair HT-PEMs that show high proton conductivity over a broad range of temperature and RH with good retention of PA. [6] Compared to PA-PBI, which shows a significant drop of proton conductivity immediately after 1st RH cycling (5–95% RH at 80 °C, Figure 3a), the RPI membrane PA-BPN1-TMA showed negligible loss of phosphoric acid leaching even after multiple RH cycles up to 95% RH at 80 °C (Figure 3b). When proton conductivity of 3rd RH cycle was plotted together (Figure 3c), RPI ion-pair HT-PEMs showed much higher value than the 3rd RH cycle data of PA-PBI due to much lower PA leaching for the latter. These newly developed ion-pair PEMs outperform Nafion in terms of proton conductivity especially at low RH and demonstrated good humidity tolerance up to 70% RH at 80 °C, suggesting the great potential for use as a membrane in HT-PEMFC without external humidification.

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<td rowspan="2">Rationale for Joint R&D Project</td>
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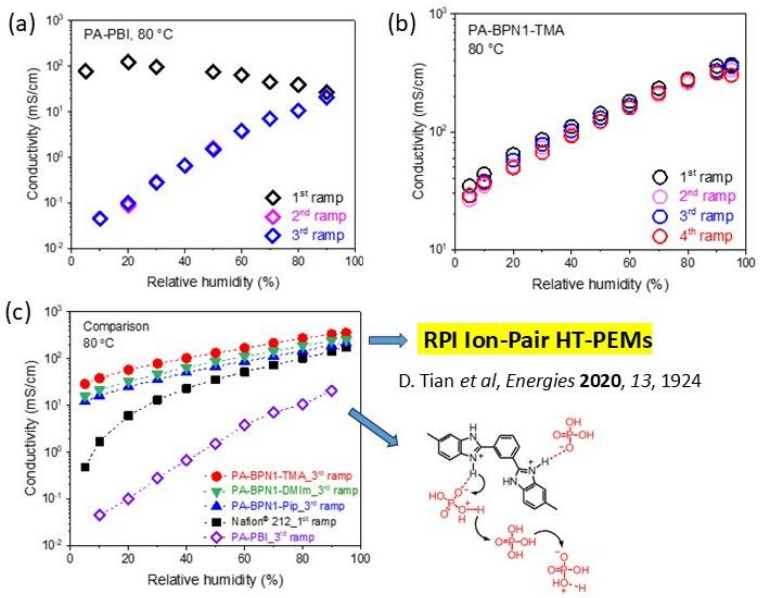

Figure 3. Comparison of proton conductivity as a function of relative humidity for Nafion, phosphoric acid-doped PBI, and RPI phosphoric acid-doped ion-pair HT-PEMs (from reference 6).
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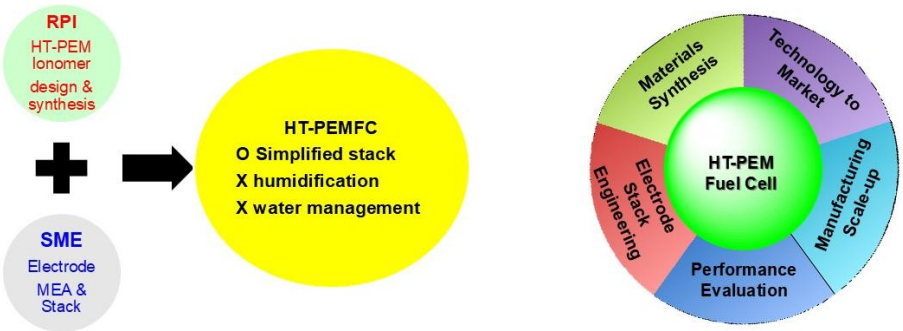
The RPI team has global reputation for high-performance PEM polymer materials development and is looking for a SME collaborator who can provide complementary expertise in MEA and stack engineering and lead the global fuel cell market. Despite the current RPI ion-pair membrane materials have shown good performance and durability in ex-situ conditions (e.g., proton conductivity), their integration into MEAs and stack level requires further optimization in design and properties of polymer materials based on feedback from SME's tests. Thus, multiple iterations of material development, cell/stack test, and feedback are critical for successful development of HT-PEMFC system. Synergistic effect of this collaboration would result in development of advanced PEM fuel cell stacks that have many attractive features and could lead global market of fuel cell technology (Figure 4).


Figure 4. Integrated collaboration approach for HT-PEM fuel cells development.
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<td>Key Words</td>
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1. hydrogen
2. fuel cell
3. proton exchange membrane
4. MEAs and stack engineering
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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 5

R&D Objectives

1. RPI: Develop and optimize ion-pair polymers for membrane and ionomer catalyst binder uses of HT-PEMFC MEAs and stack.
2. SME: Evaluate the RPI polymer materials in MEAs and stack level and suggest outcomes of materials' properties in cell tests
3. RPI and SME: Through multiple iterations of RPI materials test-and-feedback between RPI and SME, optimize materials, electrodes, and other hardware components of fuel cell stack and establish a structure–property map and guidance on scale-up.
4. SME: Demonstration of (a) HT-PEMFC performance and durability at stack level and (b) its commercial potential in fuel cell market.

R&D Target Specifications

	Performance Indicator	Unit	Target	Global Best	Standard/ Certification Criteria
1	H^+ conductivity	mS/cm	≥40	20 (Nafion)	In-plane proton conductivity at 80 °C and 50% relative humidity
2	H^+ conductivity	mS/cm	≥100	100 (PA-PBI)	In-plane proton conductivity at 150 °C and 0% relative humidity
3	% of phosphoric acid leaching	Wt %	≤10	Not available for PA-PBI	Percentage of phosphoric acid leaching amount at 3rd ramp of humidity cycling from 5% to 90% at 80 C
4	MEA performance	W/cm^2	1.3	0.73 (PA-PBI)	Max. power density in H_2/O_2 at 160 °C
6	Stack power	W	100	-	Active area: 40 cm^2, Cell No. 20
7	Continuous Max. Power Duration Time	min	5	1	Max. temperature: < 150 °C, Coolant Temp. = 35 ℃
8	Duration	hr	5,000	5,000	Accelerated Stress Test (AST) with Simulated Load Cycles (50,000 cycles)

R&D Plan

Task	Period (Month)	Key Tasks & Methodology	Main Deliverables / Milestones
Task 0	0–36	Project governance (joint meeting, reporting), IP/contracting, EHS, quality management (QC)	Kick-off meting; protocol & data management plan; periodic reports; IP strategy; final technical report.
Task 1	0–6	Basic property characterization of current generation RPI polymers materials (membrane and ionomer binder) in MEA test at SME for benchmarking. Optimization of MEA test conditions (temperature, etc).	Benchmarking report; formulation of Phase 2 plan.
Task 2	3–18	• Proton conducting polymer chemistry design: select appropriate combination of backbone and sidechain ionic group for controls of molecular weight, flexibility by tuning hard/soft backbone; try different quaternary ammonium groups and PA doping conditions. • Characterize relationships of polymer chemical structure and mechanical property, thermal	Down-selection of 1st-gen polymer candidates (≥2); structure–property map by linking chemistry variables to membrane properties and MEA test.

			property, and membrane property (IEC, PA uptake).	
	Task 3	7–24	• Development of electrodes for non-humidified HT operation & MEA optimization • Design and fabricate high-performance electrodes optimized for operation at 105–150 °C without external humidification • Integrate RPI's ion-pair ionomers into the catalyst layer to ensure high proton conductivity and robust PA retention • Implement electrode engineering techniques to achieve reduced hydrogen crossover and mitigate phosphoric acid leaching • Perform initial MEA benchmarking and provide critical property feedback to RPI for polymer material optimization	• Optimized electrode coating recipes for HT-PEMFCs. • MEA benchmarking report highlighting proton conductivity and gas barrier properties. • Established structure-property map linking electrode design to cell performance.
	Task 4	19–30	Optimizations of polymer synthetic process and membrane casting of down-selected 1st-gen candidates: molecular weight control, solvents selection, purification process, membrane casting parameter, PA doping optimization.	Down-select one polymer material candidate each for membrane and ionomer binder. Delivery of thickness-optimized down-selected PEM ≥30 pieces in 40 cm^2 size and ionomer binder solution ≥500 mL for MEA and stack tests.
	Task 5	25–36	• Large-Area MEA Manufacturing and HT-PEMFC Stack Performance & Durability Evaluation: • Establish large-area MEA manufacturing technology using down-selected RPI polymers, focusing on coating uniformity and lamination processes. • Perform stack fabrication (1 kW class) and integration based on thickness-optimized ion-pair membranes. • Execute comprehensive performance and durability evaluation using AST load cycle protocols to verify sustained power duration at 105°C and above. • Analyze heat rejection efficiency (ΔQ\ΔT) to demonstrate the potential for cooling system simplification in compact, high-power applications.	• 1 kW durable HT-PEMFC stack prototype demonstrated without external humidification. • Final report on long-term durability and degradation modes under non-humidified conditions. • Go-to-market plan and pilot production roadmap for commercialization.
	Task 6 RPI & SME	30–36	Scale up and commercialization planning: cost model, supply chain, production scale-up roadmap, and customer engagement for adoption.	Go-to-market plan; pilot production plan; partner/customer LOI (if available).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Polymer material (ion-pair membrane and ionomer) design, synthesis, and characterization: Optimize polymer architecture and synthetic chemistry to balance ion conductivity, mechanical property, and processability.
	• Advanced characterization of polymers by NMR, FT-IR, viscoelasticity (DMA/rheology), morphology, thermal property (DSC/SEM as needed).
Expected R&R of Korean SME Partner	• Electrode and stack engineering based on RPI ion-pair membrane and ionomer. • Development of electrodes for non-humidified high-temperature operation. • Development of hydrogen permeability reduction technology. • Development of large-area MEA manufacturing technology and stack fabrication. • Characterization of MEA and stack performance and long-term durability.
	• Secure intellectual property. • Lead commercialization: product spec, cost/production plan, customer expansion.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Outcomes:** • Engineering optimization of multiple materials components and manufacturing processes of HT-PEM fuel cell system: membrane, ionomer, catalysts, electrodes, catalyst coating process, GDL optimization, Bipolar plate integration. • Mapping of failure-mode of HT-PEM fuel cell MEA and stack and establishment of guideline of prevention protocols. • Verification of extended sustained power duration at peak loads. • Development of high-performance electrodes with reduced hydrogen crossover. • Demonstration of 1 KW durable HT-PEM fuel cell stack without external humidification **Impact:** • Accelerate the commercial potential of HT-PEMFC technology. • Establish a secure global supply chain for key materials of membrane and ionomer that have potential of use in global market of fuel cell through joint IP development. • Boost the global visibility and market share of Korean SMEs in fuel cell sector by creating robust international value chains. · While conventional LT-PEMFCs have a limited **continuous power duration** at peak loads due to heat rejection issues, our HT-PEMFC technology enables significantly extended **sustained power duration**.

6. References

[1] Fuel Cell Technologies Office Multi-Year Research, Development, and Demonstration Plan - Section 3.4 Fuel Cells." U.S. DOE Fuel Cell Technologies Office (2017). https://www.energy.gov/sites/prod/files/2017/05/f34/fcto_myrdd_fuel_cells.pdf

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[3] Gregory Kleen, Elliot Padgett, Durability-Adjusted Fuel Cell System Cost, DOE Hydrogen Program Record. 2021, https://www.hydrogen.energy.gov/pdfs/21001-durability-adjusted-fcs-cost.pdf

[4] Rizwan Haider, Yichan Wen, Zi-Feng Ma, David P. Wilkinson, Lei Zhang, Xianxia Yuan, Shuqin Song and Jiujun Zhang, High temperature proton exchange membrane fuel cells: progress in advanced materials and key technologies, *Chem. Soc. Rev.* **2021**, *50*, 1138.

[5] Kwan-Soo Lee, Jacob S. Spendelow, Yoong-Kee Choe, Cy Fujimoto and Yu Seung Kim, An operationally flexible fuel cell based on quaternary ammonium-biphosphate ion pairs, *Nat. Energy* **2016**, *1*, 16120.

[6] Ding Tian, Taoli Gu, Sai Nitin Yellamilli and Chulsung Bae, Phosphoric Acid-Doped Ion-Pair Coordinated PEMs with Broad Relative Humidity Tolerance, *Energies* **2020**, *13*, 1924.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization